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Release of non-methane organic compounds during simulated landfilling of aerobically pretreated municipal solid waste

Yuanyuan Zhang [a], Dongbei Yue [a,*], Jianguo Liu [a], Peng Lu [b], Ying Wang [c], Jing Liu [b], Yongfeng Nie [a]

[a] Key Laboratory for Solid Waste Management and Environment Safety, Ministry of Education of China, Tsinghua University, Beijing 100084, China
[b] Beijing Environmental Sanitation Design Science Research Institute, Beijing 100028, China
[c] Chinese Academy of Meteorological Science, Beijing 100081, China

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ABSTRACT

Characteristics of non-methane organic compounds (NMOCs) emissions during the anaerobic decomposition of untreated (APD-0) and four aerobically pretreated (APD-20, APD-39, APD-49, and APD-63) samples of municipal solid waste (MSW) were investigated in laboratory. The cumulative mass of the NMOCs of APD-20, APD-39, APD-49, and APD-63 accounted for 15%, 9%, 16%, and 15% of that of APD-0, respectively. The intensities of the NMOC emissions calculated by dividing the cumulative NMOC emissions by the quantities of organic matter removed (Q_{VS}) decreased from 4.1 mg/kg Q_{VS} for APD-0 to 0.8–3.4 mg/kg Q_{VS} for aerobically pretreated MSW. The lipid and starch contents might have significant impact on the intensity of the NMOC emissions. Alkanes dominated the NMOCs released from the aerobically pretreated MSW, while oxygenated compounds were the chief component of the NMOCs generated from untreated MSW. Aerobic pretreatment of MSW prior to landfilling reduces the organic content of the waste and the intensity of the NMOC emissions, and increases the odor threshold, thereby reducing the environmental impact of landfills.

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1. Introduction

More than 80% of municipal solid waste (MSW) generated in China with characteristics of high moisture and food waste content was disposed of in landfills in 2008. The degradation of MSW may lead to a long-term threat to the environment and public health, such as the risk of groundwater pollution due to leachate seepage, fugitive greenhouse gas emissions contributing to climate change, and odor pollution caused by non-methane organic compounds (NMOCs), which is direct harassment to adjacent urban communities. Although NMOCs comprise less than 1% (v/v) of typical landfill gas, they may create a disproportionate environmental burden and potential risk to human health (Allen et al., 1997). For example, emissions of carcinogens, such as benzene and vinyl chloride, pose a potential threat to workers and local inhabitants (El-Fadel et al., 1997; Durmusoglu et al., 2010). Furthermore, chloro- and fluoro-hydrocarbons emitted from landfills seriously corrode combustion engines that utilize landfill gas (Dernbach, 1985), deplete the ozone layer, and cause climate change (Molina and Rowland, 1974; Wallington et al., 1994). NMOCs are imposing such a significant impact on the environment that the investigation of their generation and mitigation has been receiving increasing attention (Eklund et al., 1998; Zou et al., 2003; de la Rosa et al., 2006; Chiriac et al., 2007; Kim et al., 2008; Saral et al., 2009; Raco et al., 2010).

Methods that can mitigate release of the NMOCs to the atmosphere include (1) installation of a gas recovery, collection, and treatment system, such as an activated-carbon fixed bed for NMOCs adsorption (Shin et al., 2002; Pagans et al., 2006); and (2) emplacement of cover soil with microbial activities, which has been shown to be efficient at oxidizing NMOCs. For example, some methanotrophs have been found to be able to metabolize rapidly a variety of aliphatic compounds, halogenated halocarbons, and aromatics (Scheutz et al., 2008; Tassi et al., 2009). In addition, Scheutz and Kjeldsen (2005) found that halogenated organic compounds were dechlorinated under anoxic conditions, and the generated degradation products were removed under aerobic conditions. More recently, He et al. (2010) investigated the effect of oxygen content on the release of NMOCs during the initial decomposition of MSW and indicated that high-level ventilation reduced NMOC emissions. Since aerobic degradation processes considerably decrease NMOC emissions, aerobic pretreatment of MSW may also produce desired effect on reduction of NMOCs during landfilling.

* Corresponding author. Tel./fax: +86 10 6277 3693.
E-mail address: yuedb@tsinghua.edu.cn (D. Yue).

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Aerobic pretreatment of MSW is commonly believed to be an effective method to reduce the mass of waste and improve landfill behavior. Compared to the original MSW, pollutant concentrations generated from aerobically pretreated MSW within a landfill can remarkably decrease in terms of chemical oxygen demand (COD) and nitrogen in the leachate (Leikam and Stegmann, 1999; Tränkler et al., 2002; Kuruparan et al., 2003; Bayard et al., 2008; Gunawardana et al., 2009). Besides, the biogas production potential can also be produced (Zach et al., 2000; Bockreis and Steinberg, 2005; Lornage et al., 2007). To our knowledge, however, neither has the impact of aerobic pretreatment on characteristics of NMOCs generated during anaerobic stage nor the release of NMOCs from the anaerobic degradation of aerobically pretreated MSW been reported.

The objective of this research was to characterize the NMOC emissions of aerobically pretreated MSW samples during anaerobic decomposition by measuring the NMOC yields and components and the extent of MSW degradation, and by analyzing emission intensities. The effectiveness of aerobic pretreatment of MSW in reducing NMOCs was also discussed.

2. Materials and methods

2.1. MSW sample

About 500 kg MSW sample was obtained from the Beishenshu landfill in Beijing, China. The sample was manually tumbled and mixed before the experiments. The composition of the sample obtained by manual sorting is presented in Table 1. The moisture (wet basis), volatile solid (VS), and total carbon contents of the sample were 64.5%, 65.1%, and 34.9%, respectively.

2.2. Column experiments

To estimate the amount of released NMOCs during anaerobic degradation of MSW following different aerobic pretreatments, a set of column experiments consisting of five reactors (R0–R4) was performed for 127 d. The reactors were made of high-density polyethylene. Each had an internal diameter of 500 mm and a height of 750 mm (Fig. 1).

Each reactor was filled with 77.5 kg (wet weight) of raw MSW sample at a height of 60 cm. All reactors, except for R0 which was operated anaerobically throughout the study, were initially operated aerobically at various aeration rates for 30–50 d. The normalized indicator "aerobic pretreatment degree" (APD) was calculated by dividing the amount of organic solid degraded during aerobic stage by the original amount. Accordingly, the wastes that were to be treated anaerobically were labeled as APD-0, APD-20, APD-39, APD-49 and APD-63, respectively. The operation modes are shown in Table 2. Aerobically pretreated wastes were the initial wastes for the anaerobic degradation. Subsequently, the reactors R1–R4 were operated under anaerobic conditions with efficient sealing. The accumulated NMOCs in the reactors during the anaerobic stage were analyzed on the 127th day.

2.3. Analytical methods

Moisture content was determined by heating the ground sample at 105 °C for 24 h and expressed on a wet basis. The VS content was determined by ashing the dry samples at 550 °C to the constant weight in a muffle furnace (Zhang et al., 2008). Crude protein was determined by multiplying Kjeldahl nitrogen (KN) content by the factor 6.25 (APHA, 1998). Lipid was extracted with petroleum ether for 6 h and then quantified by measuring the mass loss of the residue after combustion (AOAC, 1990, method 920.39). Crude fiber was analyzed according to AOAC (1990) method 962.09. The amount of starch was determined by subtracting the sum of lipid, protein, and fiber contents from the biologically degradable matter (BDM). The analytical procedure and calculation for BDM were the same as those described for COD (Wang et al., 2003), except that the analysis was performed at room temperature.

A 6 L SUMMA canister was used to collect the gas sample. The sample was concentrated into a cold trap which contains two beds of sorbent, i.e. Tenax TA and Carbograph, supported by quartz wool before analysis by gas chromatography coupled with flame ionization detector. The measurement setup consisted of a flow controller, a thermal desorber (Markes International, UK), and a gas chromatograph (GC6890, Agilent, USA) equipped with a flame ionization detector and jet-modulated parts (KT2001, Zeox, USA) for comprehensive two-dimensional gas chromatography (GC × GC) according to the method used by Xu et al. (2003). The sampling and desorption control software (Markes International, UK) and ChemStation (Agilent, USA) installed on a personal computer were used to control the sampling/thermal desorption system and the gas chromatograph, respectively.

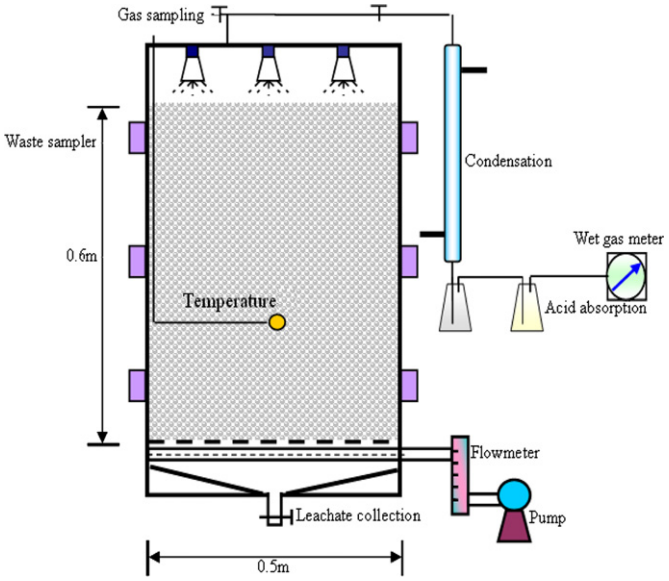


Fig. 1. Scheme of reactor changeable between aerobic and anaerobic operation.

Table 1
Composition of fresh MSW sample (% w/w).

Food waste	Paper	Textile	Wood	Plastic&Rubber	Metal	Glass	Stone	Others
62.7	12.9	3.7	1.0	5.8	0.3	1.7	3.8	7.3

Table 2
Operation modes of the five reactors.

Reactors	Aerobic pretreatment of MSW			Waste to be treated anaerobically[d]	Anaerobic
	Rate[a] (L min^{-1} m^{-3})	Frequency	Time (d)		Time (d)
R0	–	–	–	APD-0	127
R1	10	[b]	30	APD-20	97
R2	40	[c]	50	APD-39	77
R3	60	[c]	50	APD-49	77
R4	85	[c]	50	APD-63	77

[a] Gas volume flow per cubic meter of waste volume per minute.
[b] Continuously.
[c] 30-min run band 15-min break by turns.
[d] The waste to be treated anaerobically was labeled according to the aerobic treatment degree (APD). The figures in the labels indicate percentages of organic matter removed during aerobic pretreatment.

A 150 mL gaseous sample was extracted from the adsorption tube (Tenax/Carbograph 1TD/Carboxen 1000), and the sampling flow was set at 50 mL/min. The thermal desorption temperature was 300 °C. Both cold and hot jets of nitrogen gas were used. The standard mixture, EPA TO-15, was used for calibration. The jet-modulated system was operated for 6 s at a cryotrap temperature of −20 °C.

The column set used for GC × GC equipment was composed of an HP-5MS (5% phenyl, 95% dimethylpolysiloxane phase) primary column (30 m × 0.25 mm I.D. × 0.25 μm film thickness) directly coupled to a DB-WAX (polyethylene glycol phase) secondary column (1.05 m × 0.1 mm I.D. × 0.1 μm). The temperature program for the primary column began at 35 °C, holding for 8 min, and then rose to 221 °C at 3 °C/min. For the secondary column, the temperature was initially held at 30 °C for 8 min, and then raised to 210 °C at 3.3 °C/min and held for 7.5 min. The transfer line column for the system was a 1.5 m deactivated fused silica column with a 0.1 mm I.D.

3. Results and discussion

3.1. Characteristics of aerobically pretreated waste

Aerobic pretreatment produced obvious impact on the composition of the treated waste (Table 3). Higher aeration strength led to lower VS content. Moreover, easily degradable components, such as lipids, starch, and proteins, contributed to the loss of organic matter. The lipids were almost entirely removed during aerobic pretreatment. The amounts of starch were reduced by 30.4%, 40.7%, 77.7%, and 89.7% in R1 (APD-20), R2 (APD-39), R3 (APD-49), and R4 (APD-63), respectively. Protein and fiber were degraded slowly; the maximum amounts of reduction were only 28.2% and 37.0%, respectively.

The modes of aeration influenced the degradation rate of organic matter in the waste. Based on the organic losses during aerobic pretreatment of the MSW, aeration at 10 L min^{-1} m^{-3} in R1 was not adequate, and the growth of aerobic microorganisms was limited, which resulted in slow decomposition. In contrast, aeration at 60 L min^{-1} m^{-3} could supply sufficient oxygen for the degradation of organic matter, based on the decrease in the amount of starch. When the aeration rate was increased from 60 to 85 L min^{-1} m^{-3}, the efficiency of organic content reduction changed slightly, probably because excessive aeration removed more heat, which decreased the activity of the microorganisms.

3.2. Yields of NMOCs

NMOCs can be classified into five categories: oxygenated compounds (ketones, alcohols, aldehydes, acids, and esters), alkanes, alkenes, aromatics, and chlorinated compounds (Ran et al., 2009). They originate from intermediates or final products of organic matter decomposition and from organic chemicals present in MSW (e.g., household cleaning products). The majority of oxygenated compounds, alkanes, and alkenes are decomposition by-products of easily degradable organic matter (Smet et al., 1999); while most chlorinated compounds result from direct volatilization of solid waste components in the landfill, and their levels depend on vapor pressure relationships under landfill conditions (SEPA, 2004). Chlorinated compounds are produced in the largest quantities by the chemical industry for use in consumer products (Environmental Agency, 2002). Aromatics also may be present in MSW. Most aromatics are lipophilic compounds that can be absorbed by lipids (He et al., 2010). In addition, aromatics are possible by-products of lignin degradation, since lignin has an aromatic structure.

The cumulative NMOC yields during the anaerobic processes for each waste are presented in Table 4. The NMOC quantities released by APD-0 were higher than those released by aerobically pretreated waste. The cumulative weights of the NMOCs generated from APD-20, APD-39, APD-49, and APD-63 accounted for 15%, 9%, 16%, and 15% of that of APD-0, respectively. The APD-0 sample contained high starch, lipid, and protein contents. Until the 127th day, the leachate pH was stable below 5.6; this means that APD-0 was still in the hydrolysis acidification stage. Considerable acidification of organic matter inhibits the conversion of organic intermediates to CO_2 and CH_4. Hence, NMOCs, especially of biogenic origins, cannot be further degraded. Aerobic pretreatment of MSW noticeably reduced the emission of NMOCs during the anaerobic period. Besides a decrease in organic content of pretreated wastes, this phenomenon may also be attributed to the reduction in NMOCs' emissions levels of pretreated organic matter. To investigate this possibility, the emission intensity of NMOCs were calculated by dividing the cumulative NMOC emissions by the quantities of organic matter removed during the anaerobic period (Q_{VS}), which are also shown in Table 4.

The emission intensity of NMOCs decreased from 4.1 mg/kg Q_{VS} for APD-0 to 0.8–3.4 mg/kg Q_{VS} for aerobically pretreated waste. This indicates that fewer NMOCs were generated from aerobically pretreated waste than from raw waste, based on a unit weight of anaerobically degraded organic matter. Aerobic pretreatment effectively reduced the NMOC emission levels and the organic content of the waste. The reduction effectiveness, however, decreased from 82% to 20% (compared with APD-0) with the enhancement of aerobic pretreatment from APD-20 to APD-63. The compositions of the aerobically pretreated waste probably caused the differences in the NMOC emission levels. The protein and fiber contents did not vary much from APD-0 to APD-63. Compared with APD-0, APD-20 and APD-39 (from which the smallest NMOC emission levels were measured) contained relatively low lipid and starch contents, which accounted for 7.7% and 69.6% for APD-20, and 3.8% and 59.3% for APD-39, respectively. The amount of starch was further reduced to 22.2% and 10.3% with the enhancement of aerobic treatment in APD-49 and APD-63, respectively. Lipid and starch contents may have a significant influence on the intensity of NMOC emissions. Additional research is necessary to investigate this hypothesis.

3.3. Composition of NMOCs

The compositions of NMOCs generated from these wastes were investigated according to the total NMOC concentrations of each category shown in Table 5. For APD-0, oxygenated compounds

Table 3
Characteristics of waste before and after anaerobic treatment.

	Initial waste for anaerobic treatment (g/100 g dm)						Waste on 127th day	
	Starch	Protein	Lipid	Fiber	Moisture	VS	Moisture	VS
APD-0	24.3	13.1	7.8	18.4	61.8	65.1	68.0	64.2
APD-20	16.9	11.1	0.6	16.1	70.2	49.6	70.0	38.6
APD-39	14.4	10.2	0.3	14.6	70.0	45.2	70.2	41.3
APD-49	5.4	9.8	0.3	13.4	71.0	38.2	69.8	31.5
APD-63	2.5	9.4	0.1	11.6	70.0	30.2	70.4	29.6

Table 4
Cumulative mass and emission intensity of NMOCs released from fresh waste (APD-0) and aerobically pretreated waste (APD-20, APD-39, APD-49, and APD-63) during anaerobic degradation. Q_{VS} indicates the quantity of removed organic matter.

	APD-0	APD-20	APD-39	APD-49	APD-63
Cumulative mass (mg)	14.2	2.5	1.2	2.8	2.7
Emission intensity (mg/kg Q_{VS})	4.1	0.8	0.9	2.0	3.4

Table 5
Top five NMOCs released from raw and aerobically pretreated MSW samples during anaerobic degradation (μg/m^3). "Total" is the entire concentration of all detected NMOCs of each category.

	APD-0		APD-20		APD-39		APD-49		APD-63	
Aromatics										
Total		5226		1084		2810		3731		5268
1	styrene	4430	styrene	453	styrene	2379	styrene	2647	benzaldehyde	2050
2	benzene, 1-ethyl-4-methyl-	193	benzaldehyde	149	benezene	76	benezene	316	toluene	980
3	phenol	121	Phenol	126	phenol	75	toluene	159	1,4-dioxane	620
4	benezene	98	benezene	89	toluene	67	ethylbenzene	118	benezene	612
5	benzene, isoprophyl	80	1,2,4-trimethylbenzene	53	1,2,4-trimethylbenzene	30	benzaldehyde	116	ethylbenzene	244
Chlorinated										
Total		16,490		4334		1921		4888		5425
1	ethane, 1,2-dichloro-	15,589	ethane, 1,2-dichloro-	3665	ethane, 1,2-dichloro-	1717	ethane, 1,2-dichloro-	4120	ethane, 1,2-dichloro-	5177
2	carbon tetrachloride	451	carbon tetrachloride	477	carbon tetrachloride	142	carbon tetrachloride	638	carbon tetrachloride	212
3	propane, 1,2-dichloro-	269	benzene, chloro-	119	benzene, chloro-	37	benzene, chloro-	78	tetrachloroethylene	25
4	tetrachloroethylene	94	tetrachloroethylene	72	tetrachloroethylene	22	1,4-dichlorobenzene	66	trichloroethylene	10
5	benzene, chloro-	87	1,4-dichlorobenzene	16	1,4-dichlorobenzene	6	tetrachloroethylene	44	propane, 1,2-dichloro-	9
Alkanes										
Total		32,818		13,546		7272		19,885		15,921
1	pentane	16,882	pentane	10,120	pentane	3171	pentane	13,238	pentane	8547
2	1-pentene, 2-methyl-	5901	heptane	1748	heptane	2064	heptane	3316	heptane	3454
3	butane,2,3-dimethyl-	5628	octane	484	octane	1255	octane	1515	octane	1585
4	heptane	3028	hexane	456	hexane	192	hexane	581	isopentane	1178
5	octane	2094	pentane,2,2,4-trimethyl	152	isopentane	184	isopentane	496	hexane	480
Alkenes										
Total		12,346		381		391		358		519
1	limonene	10,669	camphene	129	alpha-pinene	170	camphene	170	alpha-pinene	295
2	camphene	925	alpha-pinene	109	camphene	157	alpha-pinene	67	trans-2-hexene	108
3	alpha-pinene	176	trans-2-hexene	68	limonene	23	limonene	49	camphene	54
4	trans-2-hexene	172	limonene	27	trans-2-hexene	21	trans-2-hexene	41	cis, 2-hexene	44
5	cis, 2-hexene	113	cis, 2-hexene	20	2-hexene	11	cis, 2-hexene	13	2-hexene	5
Alcohols										
Total		18,914		128		83		109		320
1	ethanol	17,157	ethanol	51	ethanol	29	ethanol	66	ethanol	293
2	isopropyl alcohol	976	isopropyl alcohol	69	isopropyl alcohol	31	isopropyl alcohol	24	isopropyl alcohol	21
3	methanol	763	isobutanol	8	isobutanol	23	isobutanol	20	isobutanol	7
4	isobutanol	18	methanol	–	methanol	–	methanol	–	methanol	–
Ester										
Total		35,537		481		4192		448		1748
1	ethyl acetate	32,498	methyl methacrylate	309	methyl methacrylate	4101	methyl methacrylate	256	methyl methacrylate	1481
2	methyl methacrylate	3039	ethyl acetate	172	ethyl acetate	92	ethyl acetate	192	ethyl acetate	267
Ketones										
Total		84,363		8132		133		571		237
1	methyl ethyl ketone	67,661	methyl isobutyl ketone	7840	acetone	61	acetone	211	acetone	118
2	acetone	12,618	acetone	158	methyl ethyl ketone	31	methyl ethyl ketone	191	methyl ethyl ketone	51
3	2-pentanone	3225	methyl ethyl ketone	68	2-pentanone	10	2-pentanone	61	methyl isobutyl ketone	23
4	2-hexanone	567	2-pentanone	26	2-hexanone	9	2-hexanone	42	2-pentanone	16
5	2-heptanone	260	2-hexanone	24	2-heptanone	8	2-heptanone	31	3-Heptanone	13
Aldehydes										
Total		6837		312		179		223		174
1	hexanal	3427	pentanal	145	undecanal	46	2-propenal	46	2-propenal	41
2	pentanal	1200	2-propenal	77	decanal	25	pentanal	44	nonanal	24
3	butanal	528	butanal	29	tridecanal	21	butanal	35	isobutenal	23
4	2-propenal	478	hexanal	15	heptanal	17	nonanal	27	pentanal	20
5	beta-pinene	290	octanal	11	2-propenal	14	hexanal	19	decanal	16

composed 66.0% of the total mass of NMOCs. For APD-20 to APD-63, alkanes were the dominant NMOCs, which accounted for approximately 48.5%, 56.5%, 66.4%, and 56.8%, respectively. The percentage of ketones decreased significantly from APD-0 to APD-63, whereas that of chlorinated and aromatic compounds increased somewhat. Oxygenated compounds, including ketones, alcohols, aldehydes, and esters, are normally observed during the initial decomposition of food waste, which coincide with the conditions of APD-0. The main oxygenated compounds for APD-0 were ketones; this observation agrees with the results of He et al. (2010).

Nagata (2003) investigated the odor threshold distribution for NMOCs. Thresholds are distributed in a wide concentration range,-depending on the odor substances. The top of the threshold distribution is 10–100 ppb for oxygenated compounds, 100 ppb to 1 ppm for hydrocarbons, and 1–10 ppm for chlorinated compounds. The order of odor threshold is oxygenated compounds < hydrocarbons < chlorinated compounds. Humans are normally more sensitive to NMOCs with lower odor thresholds. They are more sensitive to the odor from untreated waste than that from aerobically pretreated waste with the same NMOC concentration.

Table 5 also summarizes the concentration of each kind of NMOC and lists the concentration of the five most abundant NMOCs in each group. The concentrations of each kind of NMOC for aerobically pretreated wastes were all lower than that of fresh waste. For alkenes, alcohols, aldehydes, and esters, the concentration decreased by almost one order of magnitude. For

ketones, the concentration decreased by up to two orders of magnitude.

4. Conclusions

The yields and compositions of NMOCs from landfilling of aerobically pretreated MSW under laboratory conditions were investigated. The cumulative mass of NMOCs of aerobically pretreated MSW (APD-20, APD-39, APD-49 and APD-63) accounted for 15%, 9%, 16%, and 15% of that of raw MSW (APD-0), respectively. The NMOC emission levels decreased from 4.1 mg/kg Q_{VS} for APD-0 to 0.8–3.4 mg/kg Q_{VS} for aerobically pretreated MSW, and they increased as the aeration rate increased during aerobic pretreatment. Lipid and starch contents may influence the NMOC emission levels. The main categories of NMOCs for raw and aerobically pretreated MSW were oxygenated compounds and alkanes, respectively. It was found that pretreated MSW produced less intense odors. Aerobic pretreatment of MSW prior to landfilling reduces the organic content of waste, decreases the intensity of NMOC emission, and increases the odor threshold, thereby reducing the negative environmental impacts of landfills.

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